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Development of High Temperature Remedial Conformable Sand Control Systems for Use in High-Rate Gas Wells

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Abstract

Technological advances in remedial sand control and development of improved completion and downhole filter mechanisms have filled a crucial industry gap, regaining life from shut in wells. Whilst the use of a polymer-based solution, typically rated at 100°C/212°F, have proved widely popular in recent years with significant results, a challenge still exists to deploy this technology in high temperature applications, 150°C/302°F. This paper will evaluate the development process of a high temperature conformable remedial sand control system tailored to fit a high flow-rate gas well application. Laboratory testing was conducted to cover 6 main aims. To confirm temperature and fluid compatibility at 150°C/302°F to replicate field life. Acid resistance testing was also conducted to confirm short term longevity in 15% HCl to replicate acid stimulation treatments. Sand retention testing was conducted in a range of particle size distributions which is representative of the field in question, this included a mix of proppant and formation sand. Further testing such as porosity and permeability was also conducted to compare to the current polymer-based offering. Compression testing was completed to confirm the filter would be able to be compressed during deployment and return to its natural size and shape without intervention and thus able to fill all the annular gaps. Erosion testing to determine the max production rates for the filter was also conducted. Testing conducted on over 80 materials resulted in the selection of one preferred material to progress from qualification to full scale product development. Whilst not directly comparable in all aspects to that of a conventional polymer, the material was able to withstand high temperature testing (150°C/302°F) and fulfil all the aims of the project including the compression requirement. This development is completely unique to all other current remedial or primary sand control products and fills a gap within the market for high temperature conformable screen solutions. This product can regain production from shut in gas wells, it successfully retains a mixture of proppant and formation sand, offering flexibility for future stimulation activities and reductions in flaring activities.

The paper will provide an important reference for testing and design considerations of a high temperature conformable remedial sand control system tailored to fit a high flow-rate gas well application. The proposed solution will be further qualified with the aim to be deployed in a gas well that has sand production rate constraints.

Introduction

Sand production poses significant challenges in the development of gas wells, particularly those drilled into unconsolidated or weakly consolidated formations. When formation sand is produced alongside hydrocarbons, it can result in a cascade of operational problems, including erosion of downhole tubing and surface equipment, plugging of flow lines and separators, sand accumulation in wellbores, and even catastrophic well failure. Over time, uncontrolled sand production can lead to a dramatic decline in reservoir productivity, increased operational expenses, and extended periods of unplanned downtime.

Traditionally, sand control is addressed during the well's initial completion through mechanical means such as gravel packing, standalone screens, or chemical methods like resin consolidation. These strategies are designed to stabilize the formation while allowing for hydrocarbon flow. However, in many gas fields, especially older or marginal ones, sand production is often either underestimated during initial planning or arises later due to reservoir depletion, changes in drawdown pressure, or degradation of pre-installed sand control systems. As a result, a growing number of operators are turning to retrofit sand control as a remedial solution.

Retrofit sand control refers to the implementation of sand management technologies in wells that are already in production, often years after their original completion or to newly perforated zones that may have different physical properties to the previous zones. The key objective is to arrest or mitigate sand influx without necessitating a complete workover or recompletion, which can be prohibitively expensive—particularly in offshore or remote onshore locations. Retrofit solutions aim to extend the productive life of existing wells by restoring or enhancing production while maintaining well integrity and minimizing intervention-related downtime.

The technical challenges of retrofitting sand control in gas wells are considerable. Unlike oil wells, gas wells are characterized by higher flow velocities, lower densities, and more pronounced erosional effects, all of which can complicate the selection and deployment of sand control technologies. In these scenarios, advanced retrofit methods are employed, including:

- **Rigless gravel packs and cased-hole frac packs**, deployed via coiled tubing or electric-line conveyed systems.
- **Expandable metallic sand screens**, which can be deployed through tubing anchored and/or expanded in-situ to provide a mechanical sand barrier.
- **Chemical sand consolidation**, using resins or other binding agents injected into the near-wellbore formation to bond sand grains.

Selection of the appropriate retrofit strategy requires a multi-disciplinary approach involving reservoir engineers, production technologists, well intervention specialists, and geomechanics experts. Key considerations include formation properties (grain size distribution, cohesion), historical production data, reservoir pressure and temperature, previous failure modes, and logistical constraints such as well access and available infrastructure.

A nonmetallic matrix polymer solution that had been developed was identified as looking promising for use in this application. A development project was established to select and qualify a non-metallic material for use in sand retention applications. This paper will summarize the testing required to select and qualify the material for use as a retrofit sand control application.

Solution

Current technology for conformable sand control technology is limited to 100°C/212°F. To create a comparable technology for the elevated temperature range, it was necessary to develop the primary retention layer. The original technology outlined in Barclay Et al comprises a louvered base pipe, a sand retention layer, and a metal compression sleeve.



Figure 1—Conformable Sand Control Screen

The louvered base pipe enables the ID of the tubing to be maximized to prevent a restriction on production flow. The retention layer consists of an open cell matrix polymer featuring up to 80% porosity and a permeability of 43 Darcy, and it is attached to the louvered base pipe. To enable deployment the material is compressed by means of a metal sleeve to transit tight restrictions in the upper completion, once set at depth the sleeve is removed, and the material relaxes its natural size and shape completely filling the annular gap holding sand in the formation and preventing any further sand inflow.

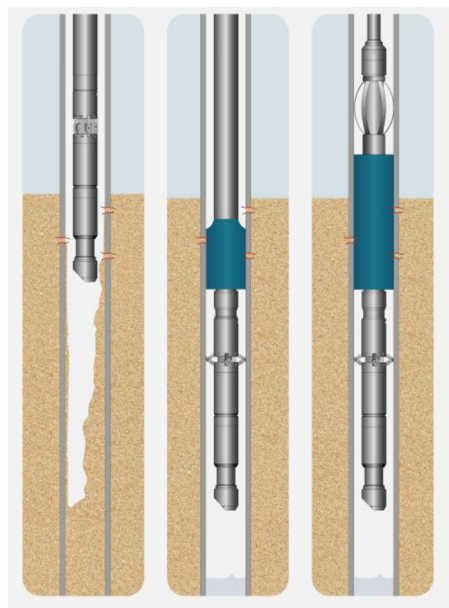


Figure 2—Tubing Deployment Sequence

The technology provides significant flexibility in deployment methods, with successful deployments carried out using slickline, coiled tubing, and pipe. The high-temperature development necessitates the use of coiled tubing or pipe for deployment.

Stage 1: The key to the technology is the use of the expandable filter, that can be deployed in hole compressed within the running tool and outer sleeve. Upon reaching the zone of interest, if required, integral jetting nozzles can be activated to begin sand clean out. The high compressibility of the polymer filter means that the assembly can pass through tubing restrictions and expand into the larger casing/liner configurations providing a conforming sand control solution.

Stage 2: The live well deployment system detailed enables sand clean-out and chemical treatments to be performed as a single trip process combined with the activation of the anchor which locks the system in place.

Stage 3: Once set, the removal of the compression sleeve enables the sand baffles to expand and conform to the ID of the casing or failed screen. The filling of the annular gap prevents further ingress of reservoir solids into the wellbore whilst still allowing passage of liquids or gases. The running tool is then pulled and retrieved from the hole.

High Temperature Development Requirements

The requirements for the new technology were as per below.

1. Temperature rating of 132°C - 150°C / 270°F –302°F
 2. To be installed in vertical/deviated and horizontal completions.
 3. To be deployed on coil tubing
 4. Allow gas production of 2-15mmscf/day
 5. Allow oil production of 100-400bpd
 6. To be set at 11,000-13,000ft depth
 7. To withstand gas /condensate and water (4-8% salinity)
 8. To cover a perforation interval of 10-100ft
 9. To be deployed through a restriction of 3.668" and set in 3.92"
 10. To hold back formation sand and proppant
1. This development was divided into five key milestones; this paper presents the work completed for milestone one. Filter Material Selection Testing
 - o Perform testing at a broad range of temperature and fluids.
 - o Test material in alternative fluids such as acids to review compatibility and limitations.
 - o Conduct sand retention tests to start building a database of performance with various sand characteristics.
 - o Porosity and Permeability of the material.
 - o Perform erosion testing with solids at temperature on the material to determine limitations.
 2. Anchor Design
 3. Sub Assembly Testing
 4. Qualification Testing
 5. Field Trial

Testing

To meet the requirements for Phase 1, the objectives listed above were necessary. A wide range of materials (>80) were evaluated and confirmed for their suitability. One material, an Organic Composite Material (OCM), currently fulfills all the goals for the new technology and will advance to Phase 2.

Fluid and Temperature Testing

Samples of the OCM material are continually tested in multiple different fluids including fresh water, crude oils, base oils, and brines at various elevated temperatures over several months. This testing has shown that the material is compatible with common well fluids anticipated to be in contact with the filter. Details of the methodology used are noted below while [Table 1](#) provides an overview of the testing results – material evaluation remains ongoing to determine longevity.

Table 1—Fluid Testing

Fluids Tested	60°C / 140°F	78°C / 172°F	93°C / 199°F	150°C / 302°F
Fresh Water	✓	✓	✓	✓
Base Oil	✓	✓	✓	✓
18% NaCl	n/a	n/a	n/a	✓

The material is cut into 2" × 2" square pieces, photographed then placed into a glass jar (<100°C) or a metal pressure vessel (≥150°C) containing the test fluid before being sealed and then put into an oven set at the required temperature. Once cooled, samples were then removed from the cell, visually inspected for change in shape and their integrity was evaluated by carefully pulling the sample by hand to check if the fluid had affected the integrity of the material. Testing to date has been performed as per [Table 1](#).

Additional testing was conducted to confirm suitability in Hydrochloric Acid (15% solution) for short term exposure (24 hours) replicating a remedial acid treatment. A 4cm square piece of the material was put into a glass jar and then 15% HCl was added until the sample was fully submerged. The sealed glass jar was subsequently positioned in the fume hood within a water bath maintained at 80°C and examined after 1, 2, 3, 4, 5, and 6 hours for any alterations in the sample's appearance and physical properties.

When checking the sample, the glass jar was removed from the water bath and the sample carefully assessed. The sample was visually inspected for change in shape and the integrity was evaluated by carefully pulling the sample with tweezers to check if the fluid had compromised the strength and form of the material.

Table 2—Acid Testing

Fluid	80°C / 176°F
15% HCl	✓

Sand Retention Testing

Multiple rounds of sand retention testing have now been completed on the OCM. Numerous samples have been carefully selected, reviewed, and assessed in both the fluids and sand laboratory to determine the quality, longevity, and suitability for the final product. This process has resulted in the proposed design consisting of three layers of the OCM leading to a total thickness of 36mm.

Sand Retention Filter Media

[Figure 3](#) shows the high magnification microscope cross-sectional image through the OCM. Retention testing has been conducted in the laboratory to fully evaluate the behaviour of the material whilst under different levels of compression to simulate the tool set inside a damaged screen or casing. Sand retention capability, pressure responses, plugging and material performance/behaviour have all been carefully assessed to find an optimum compression level in line with manufacturing requirements.

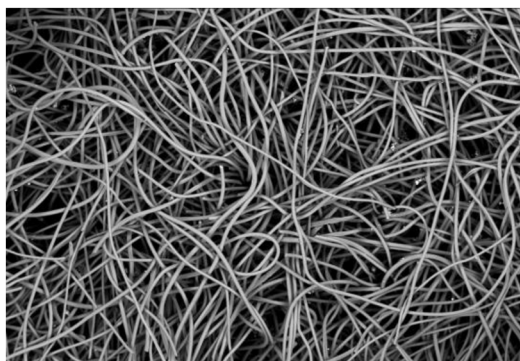


Figure 3—Material structure (x10 magnification)

Sand Retention Testing Procedure

Representative reservoir sand samples are chemically cleaned to remove any oily residue, examined under the microscope, and carefully disaggregated using a mortar and pestle (if necessary) prior to particle size measurement.

Once the particle size distribution has been measured by industry standard dry sieve analysis, the formation sand is suspended in water that has been thickened using a guar gum-based polymer to generate a stable slurry suspension.

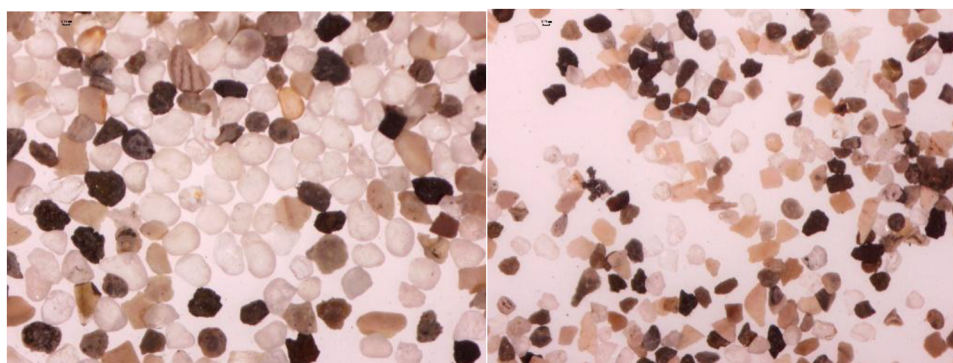


Figure 4—Fully disaggregated sand sieve fractions under high powered microscope

During the slurry tests, the prepared formation sand slurry is pumped and diluted with water before being flowed through the filter. Pressures are then recorded over time to monitor the pressure build up during the test, which then indicates if the OCM being tested is providing adequate retention and/or liable to plugging. The effluent samples collected during the test are filtered through pre-weighed 1.2mm filter membranes, which are allowed to dry and then re-weighed. The pressure data and sand through data results are collated and represented graphically for interpretation. [Figure 5](#) illustrates the test set up used.

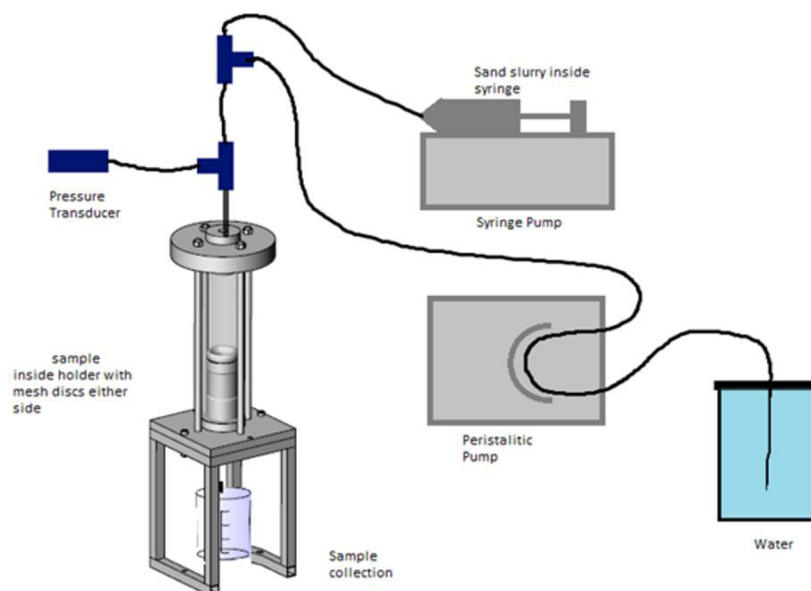


Figure 5—Sand Slurry Test Assembly

Test Sands – Reservoir and Outcrop

Reservoir sands are typically used for retention testing and are always preferred when recommending sand control solutions. There can be some variations in testing results between using reservoir and outcrop sands due to the difference in particle shape and mineralogy of the grains. Outcrop sands tend to be well rounded and ground silica is very angular whereas reservoir sands are composed of a variety of minerals and clays, governing a difference in shape properties. SPE-178966-MS provides more detail. However, if there are no reservoir sands available, outcrop sands are the next best alternative and can still provide a good understanding of performance and reliability.

The initial material qualification programme for the technology used several different outcrop sands; these sands were combined in unique proportions to generate multiple different individual test sand mixes. The particle size distributions of each test sand were measured by dry sieve analysis and a comprehensive programme of retention testing performed to gain an in depth understanding of the polymer performance. Focus was placed on poorly sorted sands with a smaller d_{10} and high fines content. These are the types of challenging sands expected for remedial sand control investigation where the technology may be utilised. Several reservoir sands have since been included as part of the development programme to assess behaviour and performance of the OCM. Figure 6 and Table 3 show the particle size distributions of a few of the sands tested.

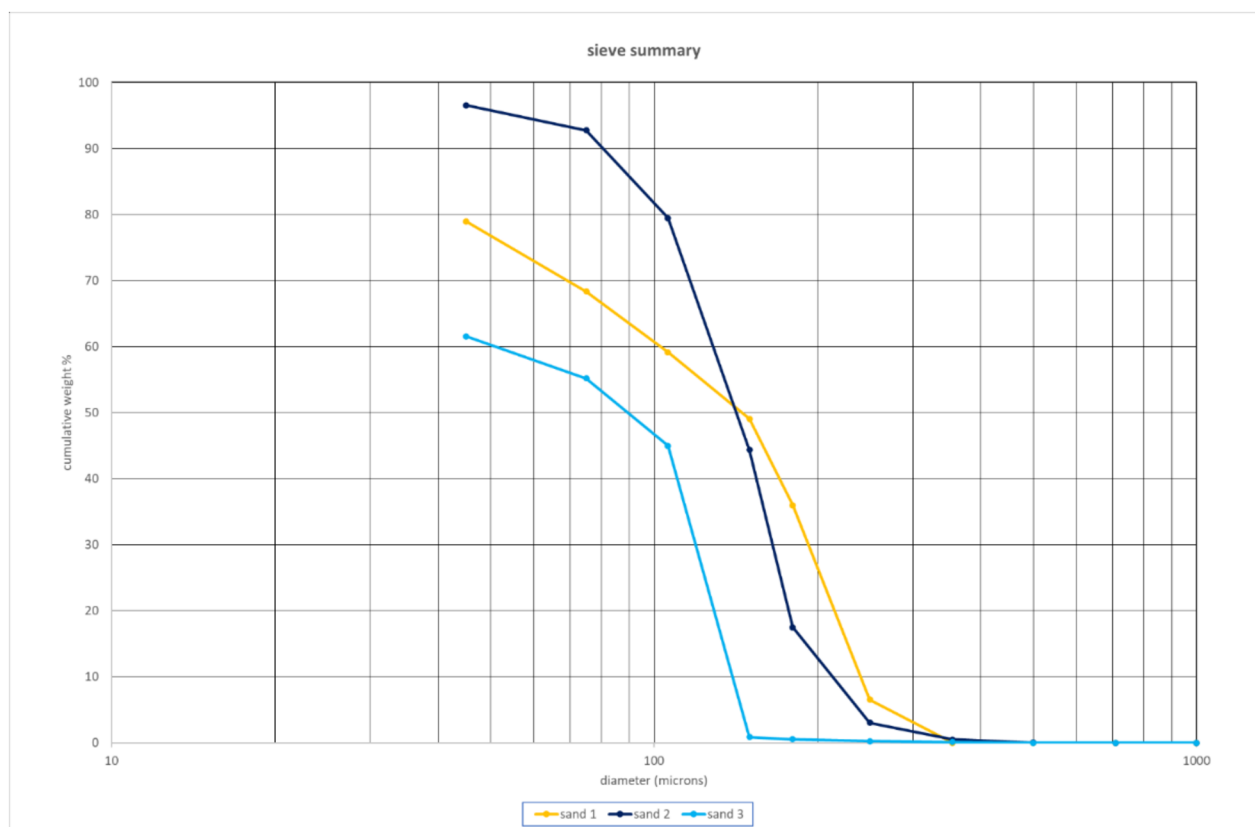


Figure 6—Test Sand Particle Size Distribution

Table 3—Test sand particle size parameters

Sand	d5	d10	d50	d90	Fines (%)
Sand 1	275	242	145	<45	21
Sand 2	240	216	143	82	2.6
Sand 3	214	177	77	<45	>30%

High Temperature Test Sand

A single sand sample as chosen from a comprehensive internal database that most accurately represented sands similar to those in the selected field. The selected sand has a d50 value of c. 236 microns and contains 2.7% fines (particles smaller than 45 microns).

Particle Size Data

The particle size distribution for the test sand used is plotted in Figure 7. The sieve data graph shows that the sample is uniform and well sorted with a d50 value of c. 236 microns and c. 2.7% fines (<45 microns) although this value would most likely be higher if measured by laser particle size analysis (LPSA). Figure 8 presents the provided data for the application with the test sand overlayed. The test sand chosen is marginally finer than the customer data but is deemed representative of the sands likely to be encountered in the field.

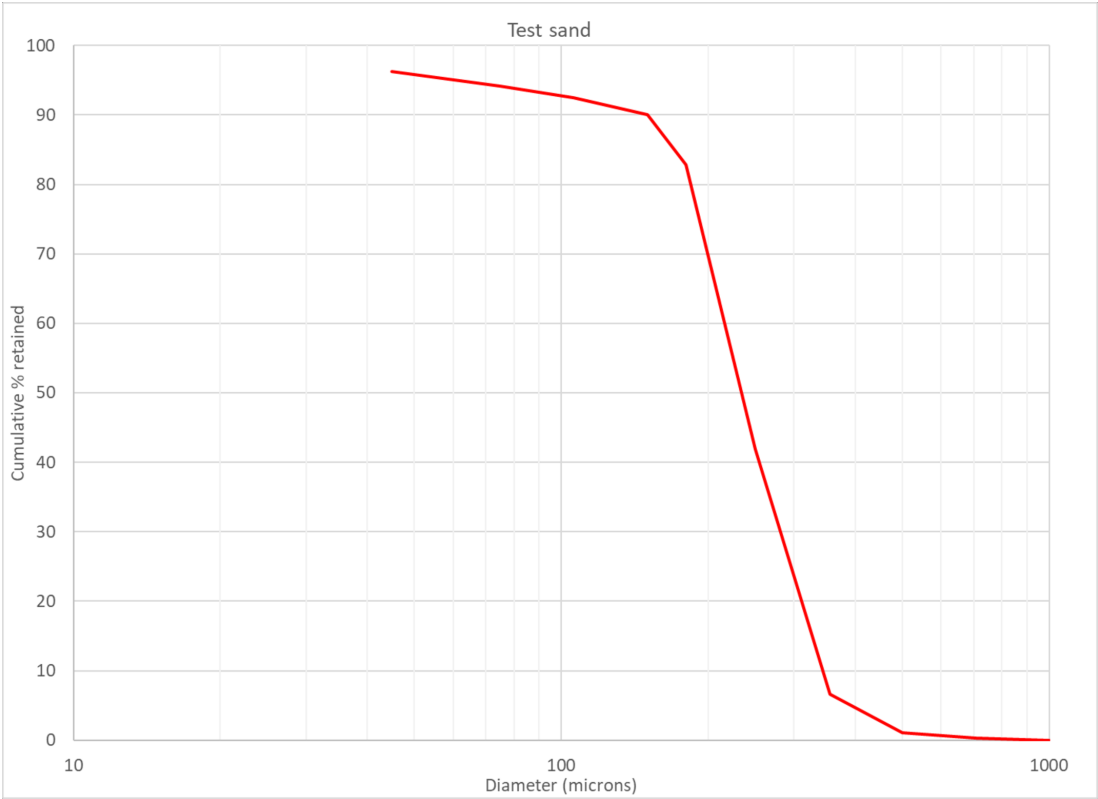


Figure 7—Sieve Data for Test Sand

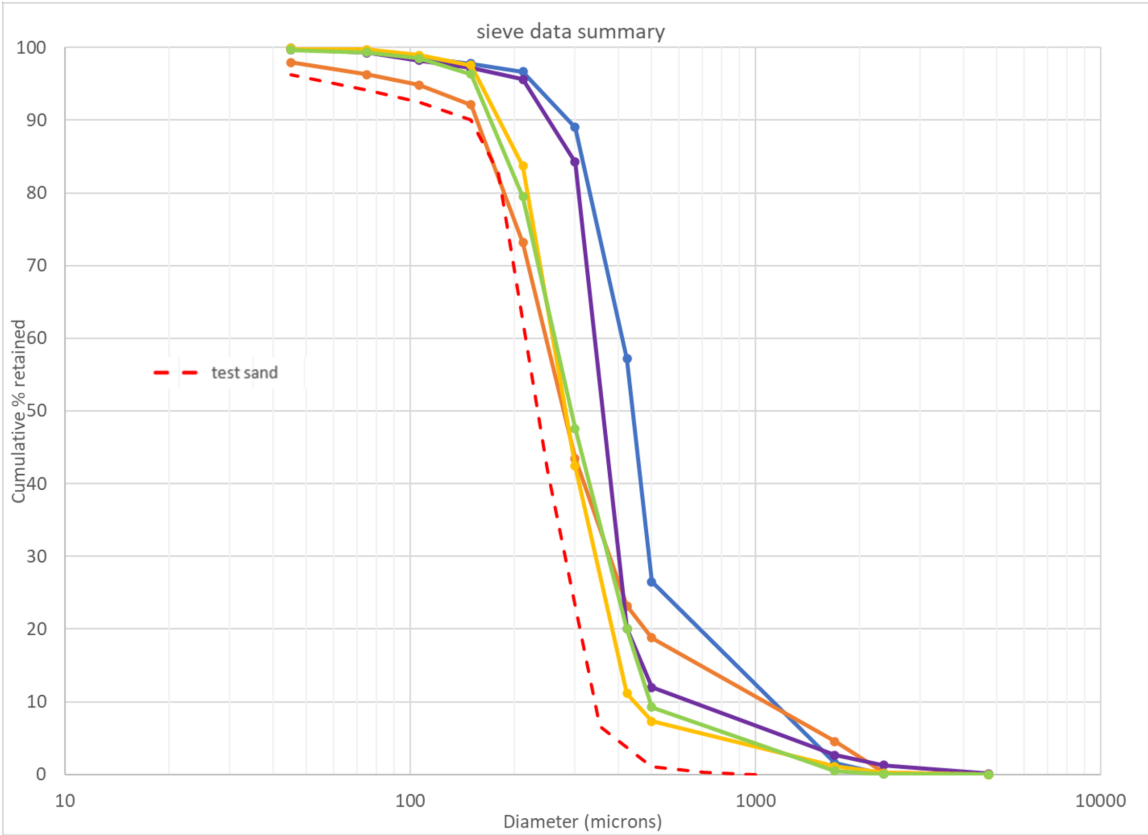


Figure 8—Produced Sand Summary

Slurry Testing

It should be noted that these retention tests examine overall trends with filter performance rather than detailed data analysis and are not replicable to actual field data/rates. The total amount of sand passing through the material in each test performed is shown in [Table 4](#). In the tests, the OCM was layered and compressed to replicate requirements for the customer application.

Table 4—Total sand passing through material in all tests

Sand	OCM Configuration	lb/ft2	grams
Test Sand	Multi-Layer, Compressed	0.044	0.151
Customer produced sand	Multi-Layer, Compressed	0.007	0.023

To further qualify retention performance of the material, it is beneficial to compare the change in sand through with time and pressure profile. The sand through data should quickly fall to a low level during the test if sand control is achieved; if control is poor, then significant amounts of sand will continue to be produced well into the test. The pressure drop observed will be a combination of the sand permeability and any plugging of the filter – the filters themselves are too permeable to generate a significant pressure drop. Therefore, the pressure profiles for a particular sand should be similar on each filter given good retention and no plugging. At the end of the retention test, sand injection is stopped to perform a swab and surge style test to help identify any plugging with the filter. The flow of clean water is started and stopped at regular intervals and the pressure observed for any inconsistencies.

[Figures 9](#) and [10](#) show the test results obtained for multi-layer compressed OCM using the chosen lab test sand showing encouraging results. The cumulative sand through displayed in [Figure 9](#) demonstrates low levels of sand production during the test. The pressure profile recorded is low and stable matching the sand characteristics. Although these results show slightly more sand through than has been seen with previous material developmental testing there are no concerns. There is no evidence from this test that plugging is an issue either.

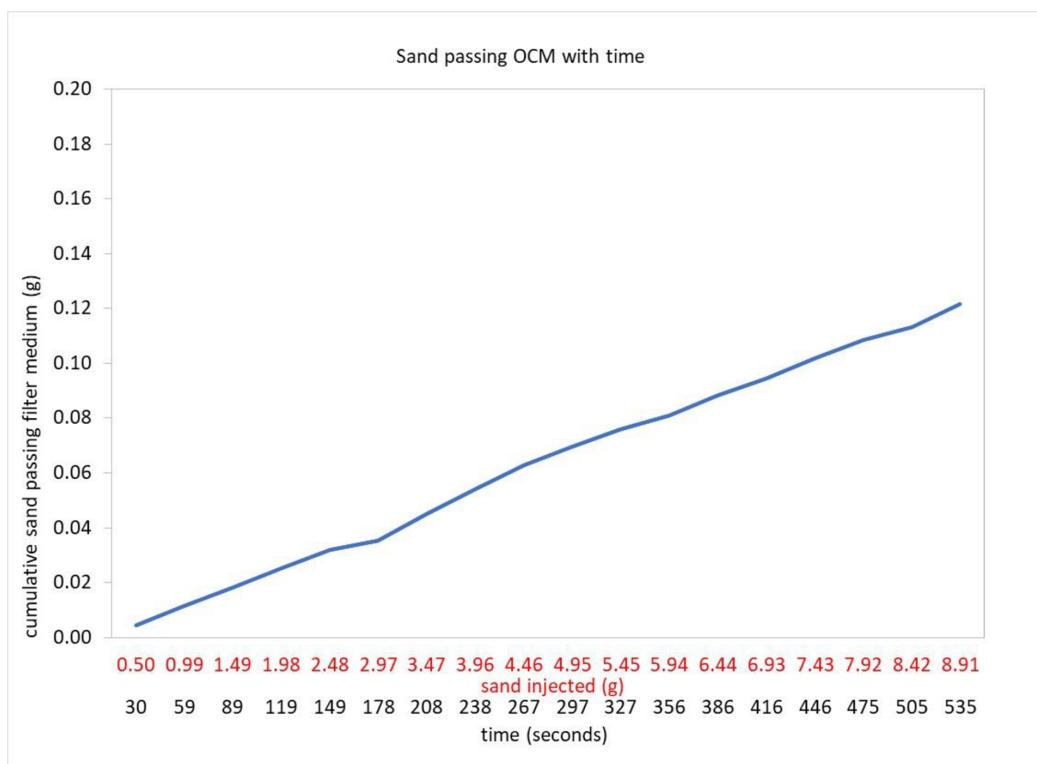


Figure 9—Cumulative sand through with time in the test (multi-layer compressed OCM, lab test sand)

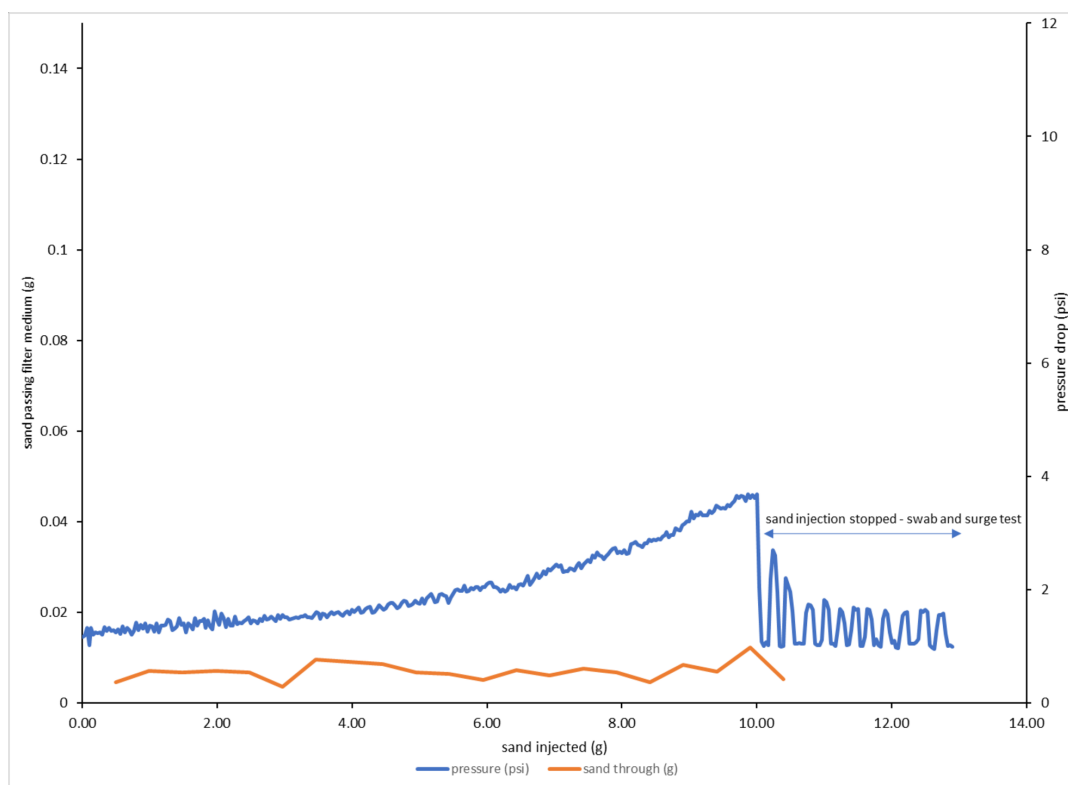


Figure 10—Pressure Profile – with swab and surge test and sand through with time (multi-layer compressed OCM, lab test sand)

Slurry testing using reservoir sands

Additional testing was performed using a single produced sand from the chosen field and provided by the customer, as detailed in Figure 11. The sand was chemically cleaned and placed through a range of sieves to remove the larger proppant grains >500 microns. Using this sand sample, retention testing was performed using the multi-layer compressed configuration OCM. The pressure profile recorded and cumulative sand through recorded during the test can be found below.

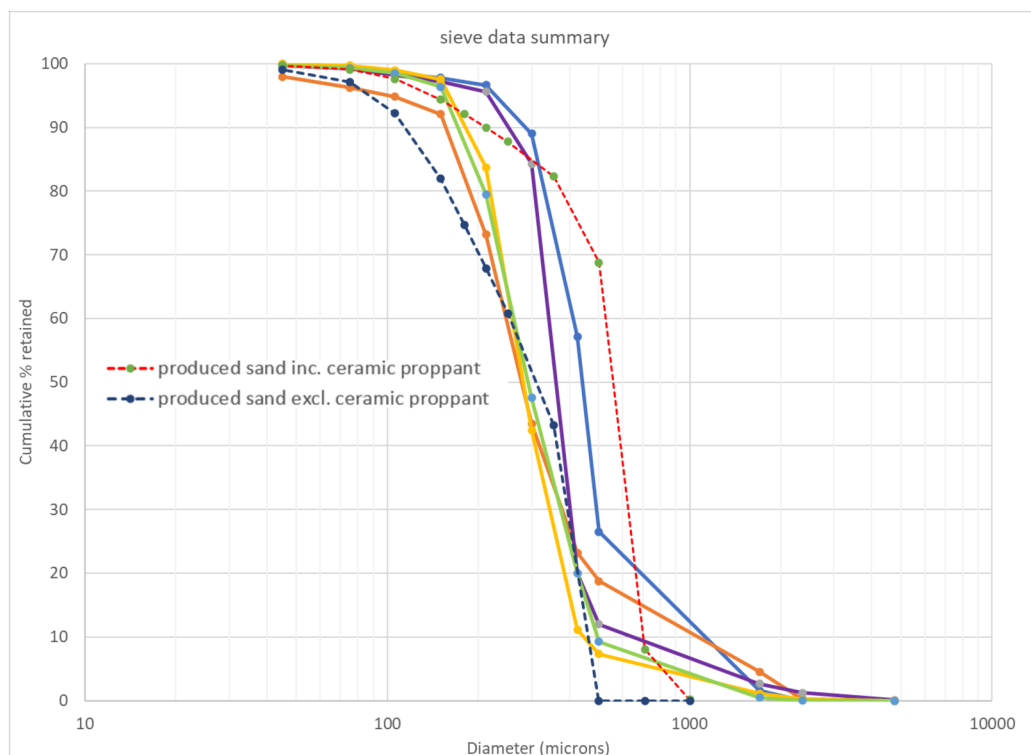


Figure 11—Original sieve data for application and customer provided produced sand.

Figures 12 and 13 present the test result using the produced sand sample, showing positive results. The cumulative sand through shown in Figure 12 shows very low levels of sand production and improved retention compared to the first test using the lab test sand; and the pressure profile recorded during the testing in Figure 13 reflects this. There is no evidence to suggest any plugging is occurring with the produced sand and the material; reliable sand retention is achieved.

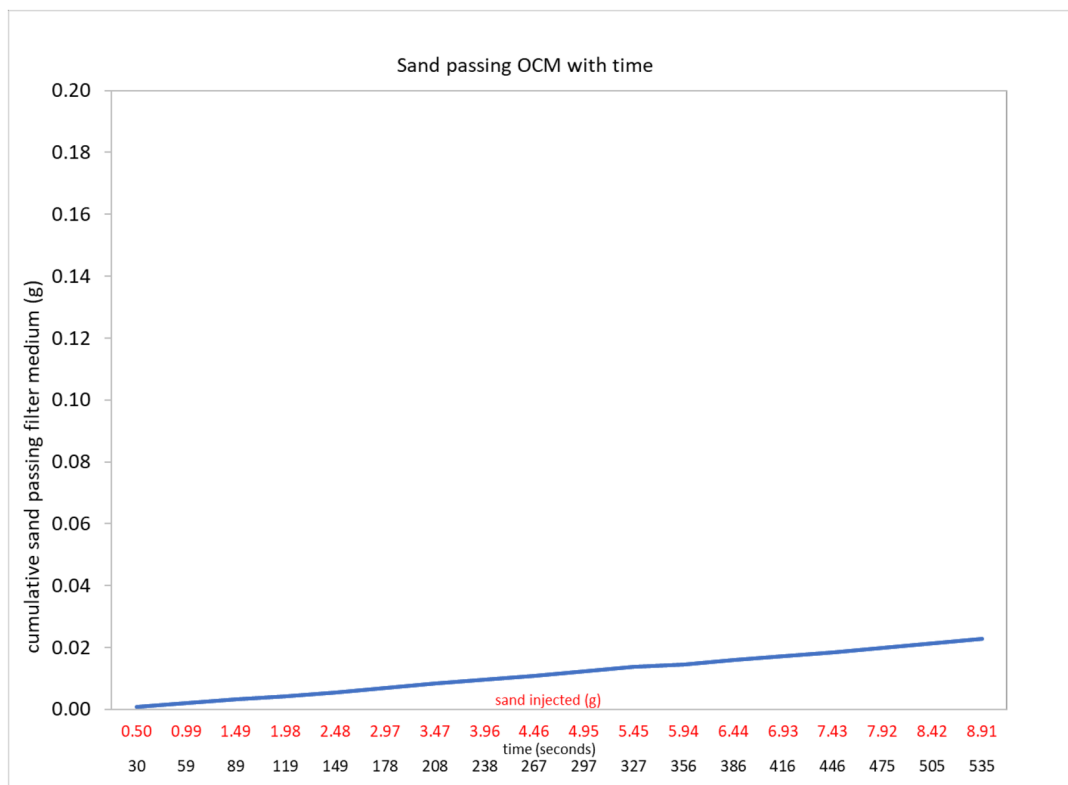


Figure 12—Cumulative sand through with time in the test (multi-layer compressed OCM, produced sand)

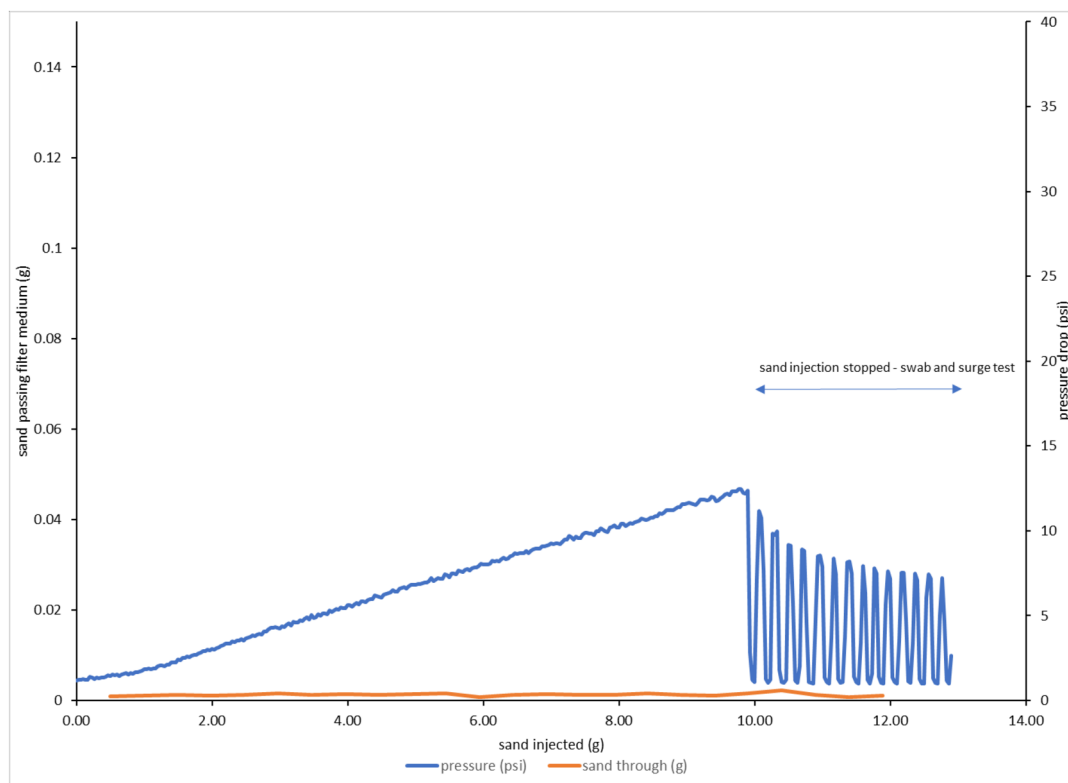


Figure 13—Pressure Profile – with swab and surge test and sand through reducing through time (multi-layer compressed OCM, produced sand)

Additional Measurements

To gain a deeper insight into the performance of the customer produced sand with the OCM, several calculations were performed using the total amount of sand injected during the test and the weights of sand that passed through the filter media as well as that which was retained on its surface. Less than 0.3% of the injected sand was found in the effluent samples collected from the OCM, while more than 60% of the sand remained on top of the material after testing. Microscope images illustrate that sand is retained within the material after testing, which aligns with the pressure results recorded.

Filter Retention Mechanism

These filters have been evaluated in a cylindrical form and placed into custom-designed holders. After the test concludes, the cylinders are carefully extracted from the holder, and photographs are taken of the top and bottom surfaces. The cylinder is then cut axially, and images of the cut face are captured at the top, centre, and bottom. These images provide a clear indication of the sand distribution within the filter at the end of the test, although some movement of sand grains is anticipated due to the sectioning process. As would be expected, a layer of sand forms at the top surface and the invasion depth depends at least in part on the size of the sand grains.

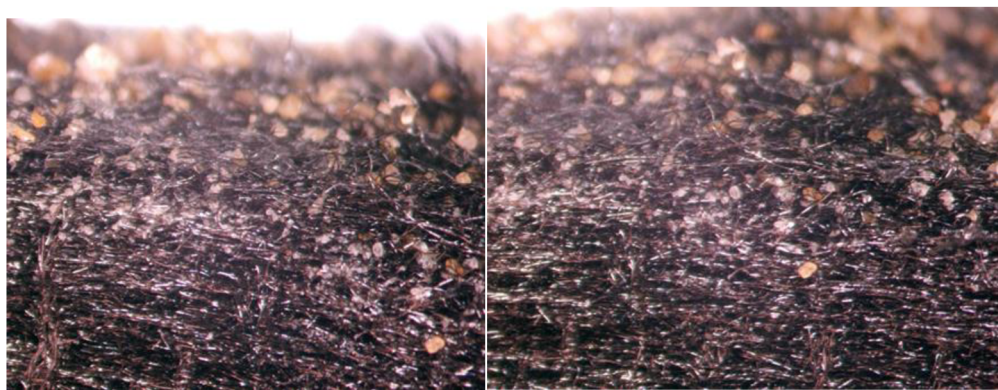


Figure 14—Sample after produced sand test demonstrating sand ingress

Porosity and Permeability

Porosity

The porosity of the material has roughly been estimated to be 59% by the procedure below.

1. A piece of material was cut to fill a 50ml measuring cylinder. The OCM was soaked in water and then pushed into the cylinder, which was already full of water.
2. The material was moved around and compressed to remove as much trapped air as possible before being pulled back to its original size. Any excess water above the OCM was removed and the volume noted as VT.
3. Next, all the water was poured out into a beaker with the OCM then carefully removed and squeezed to get all the water out without losing any.
4. Finally, the water in the beaker was poured back into the measuring cylinder and the volume noted as VR.

The % porosity was calculated by dividing the volume of water left once the OCM was removed (VR) by the total volume occupied by the OCM and water (VT) – see below.

$$\begin{aligned} \text{VR/VT} \times 100 &= \% \text{Porosity} \\ 17/29 \times 100 &= 58.6\% \text{Porosity} \end{aligned}$$

Permeability:

An indication of the retained permeability within the material once the sand retention test had been completed – post sand injection. These measurements have been recorded based on fixed flow rate before and after inclusion of the sample and calculated based on the increased pressure using Darcy's equation below.

$$K = Q\mu L / A\Delta P$$

where:

k= permeability

Q= flow rate

μ = fluid viscosity

L= length

A= cross sectional area of flow

ΔP = increased pressure drop

Table 5—Initial and retained permeability measurements in Darcies for produced sand, multi-layer compressed OCM (D)

Initial Permeability of material	31.65
Retained Permeability with sand layer on top	3.401
Retained Permeability with sand layer removed	22.79

Once the retention test had been completed, the layer of built-up sand was carefully removed from the top of the material, and the retained permeability re-measured. Although there does appear to be a significant reduction in permeability of the media post sand retention, the pressure drop observed is partially attributable to the layer of sand built up on the surface of the material and doesn't necessarily represent a reduction in permeability of the material itself.

Erosion

The objective of this test was to determine an erosion rate for the material in liquids containing solids to enable the service life for a given set of well conditions and to allow comparisons with conventional screen constructions.

Method

The material was secured onto a louvered base pipe to replicate full scale manufacture. The coupon sample was then carefully fitted inside the sample support and installed into the test jig. An orifice replicating a perforation of 0.5" was used on the test to simulate flow through perforation onto the material.



Figure 15—Erosion test sample

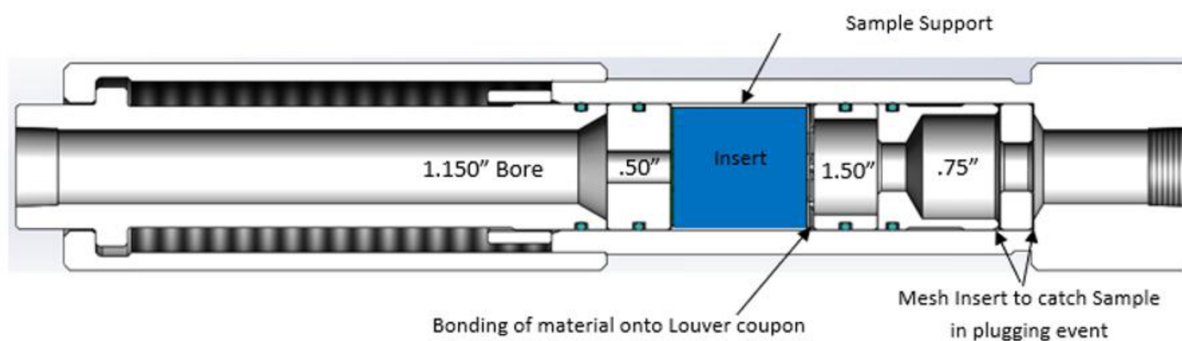


Figure 16—Test Setup Erosion Testing

Pre-Test Information – Orifice Velocities and Flow Rate:

The fluid speed through the nozzle is approximated using the equation: -

$$Q = v \cdot A$$

The area of the 0.500" orifices is 0.1964in². See [Table 6](#) below for target velocities v's flow rate.

Table 6—Flow Rates through the orifice

Flow Rate (L/hr)	Flow Rate (L/min)	Orifice - .500" (ft/sec)
100	1.67	.72
200	3.33	1.44
300	5.00	2.16
400	6.67	2.88
500	8.33	3.60
600	10.00	4.32
700	11.67	5.04
800	13.33	5.76
900	15.00	6.48
1000	16.67	7.19

The sand loading used was 0.2% solids by weight. Based on a total loop volume of 200L (200,000 cm³) this gives a sand mass of 400g. ($200,000 \times .002$). Sand used was sieved to be less than 45 μ m.

1. Clean the Fixture re-assemble.
2. Circulate the flow loop with water starting at 200 L/hr flow rate. Once the flow has stabilised, gradually pour sand into the tank. Once the sand is in the tank, stir the mixture to distribute the sand evenly.
3. Circulate mixture for 5 minutes. Monitor the upstream P (in) and dP pressures and measure actual flow rate using a stopwatch and 1 litre container. Keep fluid sample for sand concentration checks.
4. Increase pump rate at 100 L/hr flow rate increments. The increased flow rate should be done gradually. If the P (in) or dP pressures starts to rise at an increasing rate allow to stabilize before increasing further. Initially these steps should be held for 5 mins or until the pressures are stable.
5. At 500 L/hr maintain the flow rate for 15mins.
6. Continue to increase the flow rate in 100 L/hr steps allowing each step at least 15mins for a stable pressure condition to be observed. If the dP pressure shows signs of gradual increase allow more time to stabilise. If the dP pressure starts to rise at an increasing rate reduce the flow rate gradually to the previous stable condition.
7. Once the maximum flow rate has been established maintain that condition for a minimum of 4 hrs.
8. Before reducing the flow rate and stopping the pump take another 1L sample of the fluid for checking the sand concentration.
9. Once the pump has been stopped, disassemble the fixture and visually inspect the filter. Document observations and findings with photos.

A stabilised pressure was built quickly in the test, settling at roughly 3.5 bar up to a flow rate of 300 l/hr (2.16 ft/sec). As soon as the flow rate was increased beyond that the differential pressure rapidly increased to 8.5 bar. The rate was dropped to 100 l/hr (0.72 ft/sec) for a couple of minutes before increasing to 200 l/hr (1.44 ft/sec). The flow rate was maintained for four hours. Over the course of the hold, the differential pressure increased from 2.6 bar to 6.5 bar.

After the four hours the flow rate was increased once more to confirm we had reached the max flow rate. The rate was increased to 300 l/hr (2.16 ft/sec), and the pressure increased to 10 bar. Over 20 minutes the pressure increased to 12 bar and looked to keep increasing rapidly. It was deemed that the maximum flow rate achievable was 200 l/hr (1.44 ft/sec). The results of the test are shown in Figure 1741.

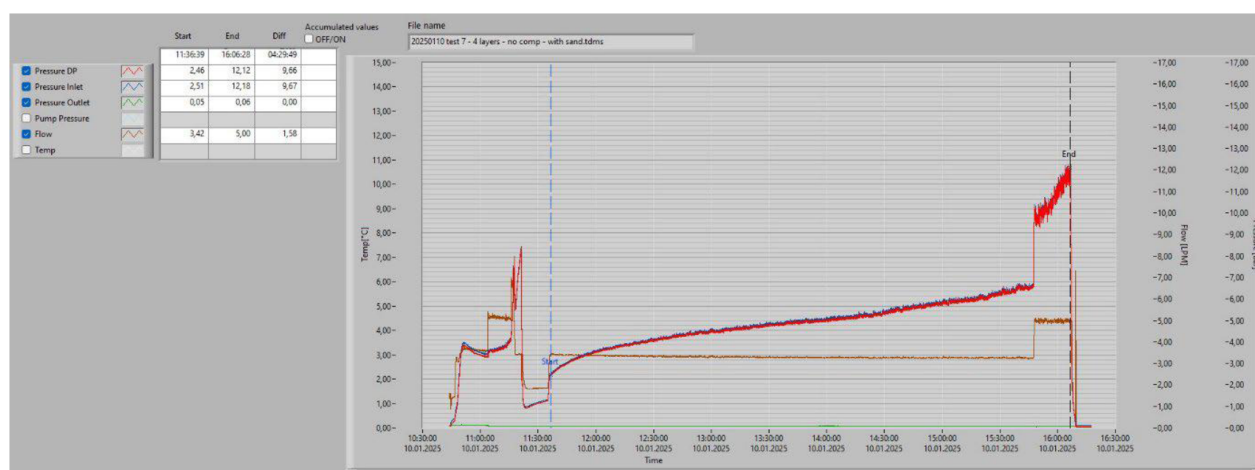


Figure 17—Erosion Test Chart

The test setup and fixture were disassembled, the sample was recovered, Figure 18 shows the sample after the test. This sample showed that it had no signs of erosion on its top surface.



Figure 18—Erosion Test Sample after test

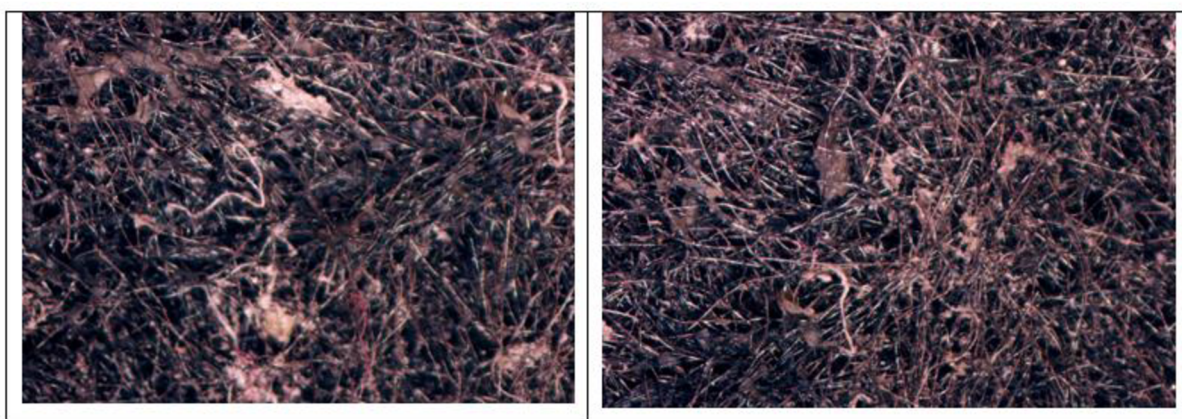


Figure 19—Test sample after erosion testing

Due to the arrangement of the fibers of the material, the flow rates and velocities seen during the tests were lower than what had been seen in previous material erosion tests. This makeup played a significant role in attempting to achieve stable differential pressure, especially in the sand loaded erosion flow tests. The fine particles, the nature of the materials makeup, and additional compression due to the kinetic energy from the jet of fluid into the material made it susceptible to plugging.

Although the flow rates were not as high as previous tests, there was no evidence of erosion in the sample. 1.44ft/sec was recorded when tested with sand, this was not a limitation of the erosion rate of the material but rather a limitation on the test setup where a small sample (50mm diameter) was used and exposed to the rates. The sand used was all fines which in application would be bridged by larger particle sizes so would not plug the material. This has also been noted in numerous sand retention testing where the material does not plug.

The rates seen without sand exceeded 7ft/sec with no damage to the material. Standard metal screens are typically reduced to less than 1ft/sec rates to prevent erosion, this testing has shown that the material can exceed these rates without showing any signs of erosion.

Conclusion

Preliminary testing has demonstrated success in creating a high-temperature, conformable, nonmetallic sand retention medium suitable for high-temperature gas wells. The Organic Composite Material (OCM)

effectively passed the criteria for Stage 1 which involved testing for temperature and fluid compatibility, acid testing, porosity and permeability, sand retention testing and erosion testing. The next stage of the project reviews the mechanical hardware of the technology prior to moving to sub assembly manufacturing trials and ultimately full qualification of the technology. This development represents a step change in conformable sand control technology with the product being able to regain production from shut in gas wells and reduce the carbon intensity of accessing reserves. It can successfully retain a mixture of proppant and formation sand and offering flexibility for future stimulation activities and reductions in flaring activities.

Nomenclature

OCM Organic Composite Material

References

- Barclay, Amy, McKay, Eilidh, and Suzanne Stewart. "Novel Thru-Tubing Sand Control Solution: Defining the Operating Envelope." Paper presented at the SPE Asia Pacific Oil & Gas Conference and Exhibition, Virtual, November 2020. doi: <https://doi.org/10.2118/202300-MS>